SoDA 501: Experiments

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Notation

- Z_i: assignment (randomized indicator)
- D_i: treatment received (exposure / compliance)
- Y_i: outcome (clicks, purchases, retention)
- X_i: pre-treatment covariates (e.g., baseline activity, blocks)

Average Treatment Effect (ATE)

Definition

$$ATE = \mathbb{E}[Y(1) - Y(0)]$$

In a randomized experiment with full compliance ($D = Z$)

$$\widehat{ATE} = \bar{Y}_{Z=1} - \bar{Y}_{Z=0}$$

When to use

- Full compliance or treatment is perfectly delivered
- You want the population-average causal effect

Intent-to-Treat (ITT)

Definition

$$ITT = \mathbb{E}[Y \mid Z = 1] - \mathbb{E}[Y \mid Z = 0]$$

Interpretation

Effect of being *assigned* to treatment, regardless of whether it is received.

When to use

- Non-compliance exists (not everyone receives the treatment)
- Product/policy decisions are made at the assignment level

TOT / LATE (Instrumental Variables)

Definition (conceptual estimand)

$$TOT = \mathbb{E}[\, Y(1) - Y(0) \mid D = 1\,]$$

Key issue: D (treatment received) is often *not randomized*, so $\mathbb{E}[Y \mid D = 1] - \mathbb{E}[Y \mid D = 0]$ is generally biased.

With non-compliance, D is not randomized. Use Z as an instrument for D.

Wald estimand (LATE)

$$LATE = \frac{\mathbb{E}[Y \mid Z = 1] - \mathbb{E}[Y \mid Z = 0]}{\mathbb{E}[D \mid Z = 1] - \mathbb{E}[D \mid Z = 0]} = \frac{ITT}{\text{first-stage}}$$

Interpretation

Average causal effect for *compliers* (units whose treatment receipt changes with assignment).

IV Assumptions for LATE

1. **Relevance:** Z affects D (first-stage $\neq 0$)
2. **Exclusion:** Z affects Y only through D
3. **Independence:** Z is randomized
4. **Monotonicity:** no defiers

Estimation in R

Diff-in-means (ATE/ITT)

```r
ate_itt <- with(df, mean(Y[Z == 1]) - mean(Y[Z == 0]))
```

Regression adjustment (cluster-robust SE)

```r
library(estimatr)
fit <- lm_robust(Y ~ Z + X1 + X2 + factor(block),
                 data = df, clusters = cluster_id)
summary(fit)
```

IV / LATE (2SLS)

```r
iv <- iv_robust(Y ~ D + factor(block) | Z + factor(block),
                data = df, clusters = cluster_id)
summary(iv)
```

Estimation in Python

Diff-in-means

```
ate_itt = df.loc[df.Z==1, Y].mean() - df.loc[df.Z==0, Y].mean()
```

OLS with block fixed effects (cluster SE)

```
import statsmodels.formula.api as smf
fit = smf.ols(Y ~ Z + X1 + X2 + C(block), data=df).fit(
    cov_type=cluster, cov_kwds={groups: df[cluster_id]}
)
print(fit.summary())
```

IV / LATE (2SLS)

```
from linearmodels.iv import IV2SLS
iv = IV2SLS.from_formula(Y ~ 1 + C(block) + [D ~ Z], data=df).fit()
print(iv.summary)
```

Multiple Testing and FDR

When testing many outcomes, some will look significant by chance.

False Discovery Rate (FDR)

Expected share of false positives among rejected hypotheses.

BH (Benjamini–Hochberg) controls FDR at level q (often 0.05).

BH Procedure

Sort p-values:

$$p_{(1)} \leq p_{(2)} \leq \cdots \leq p_{(m)}$$

Find the largest k such that:

$$p_{(k)} \leq \frac{k}{m} q$$

Reject hypotheses corresponding to $p_{(1)}, \ldots, p_{(k)}$.

BH in Code

R

```r
p_adj <- p.adjust(pvals, method = BH)
```

Python

```python
from statsmodels.stats.multitest import multipletests
rej, p_adj, _, _ = multipletests(pvals, alpha=0.05, method=fdr_bh)
```

Practical Guidance

- Report **ITT** for product/policy decisions (assignment-level effect).
- Estimate **LATE** if non-compliance is meaningful and assumptions are plausible.
- Always check first-stage strength ($Z \rightarrow D$).
- Use block fixed effects if you randomized within blocks.
- Use BH/FDR when you test many outcomes.

Summary Table

Quantity	What it measures	Typical use
ATE	Population average effect	Full compliance
ITT	Assignment effect	Default in experiments
LATE	Effect for compliers	Non-compliance + IV
BH/FDR	Controls false discoveries	Many outcomes/tests